

Article

The Effectiveness of Tannin on the Amount of Damage to Forest Trees and Stands Caused by Red Deer in the Western Carpathians

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Abstract: Damage caused by cloven-hoofed game and wild animals is a serious problem in Slovakia and neighboring countries. In 2022, the damage to forests in Slovakia was estimated at a level of almost EUR 2.9 million. Despite several implemented measures, the situation is so severe that successfully restoring some trees in some heavily attacked stands is impossible. This paper deals with the preventive elimination of damage caused by red deer in forest stands by the biological method of applying tannin-based mineral licks. Specifically, licks containing tannin obtained from the edible chestnut were used for testing. Several positive effects are attributed to tannins in developing livestock and animal fitness status. To date, however, the question of using tannins as biological agents for the possible protection of forest stands has not been addressed. Therefore, red deer damage was assessed at two localities with similar natural conditions in the central part of the Western Carpathians to explore such a possibility. After the initial sample measurement of the bark-stripping damage caused by the red deer in the autumn, mineral licks with tannin were placed in one selected study area over the winter. The levels of and differences in forest damage by deer between areas with the application of licks with tannin and without tannin were detected in spring of the following year. In the location treated with tannin, the number of damaged sample plots decreased, and the mean plot damage did not increase (no statistically significant differences were registered in mean plot damage between autumn and spring). Meanwhile, in the control location, the total amount of bark-stripped area doubled, mainly due to a statistically significant increase in mean plot damage between autumn and spring. Based on this, we conclude that tannin had a significant preventive effect and considerably reduced the extent of bark stripping. Such promising results indicate that tannin licks can effectively mitigate the problem of bark-stripping damage in areas heavily attacked by red deer.

Keywords: forest protection methods; deer damage; bark stripping; tannin licks



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1. Introduction

Forestry's priority mission is to maintain and promote the provisioning, regulative, and cultural services of the forest. The essential tool for ensuring them is different intentional management activities, among which we can also include integrated forest protection against the harmful effects of wild animals. There is a mostly positive relationship between the forest ecosystem and wildlife based on trophic and topical links, but because of economic concerns and some wild animals overlapping, some undesirable forest damage can appear, and forest managers need to find a proper management solution.

Due to climate change's negative impacts, forest stands have been damaged by a larger spectrum of biotic, abiotic, and anthropogenic harmful agents in recent years [1]. Among

the more important ones in Slovakia, cloven-hoofed game and wild boars were registered; these animals disrupt the forest's natural and artificial regeneration processes, making it impossible to implement the currently promoted close-to-nature management in forests [2]. In general, the damage very negatively affects the health status and vitality of the trees and whole stands, as well as reducing the ability of trees to withstand droughts, fungi, and insect infestations. Moreover, the commercial value of the trees is permanently reduced. The need for protection is acute, especially considering the trees' diminished adaptability.

The intensity of animal damage in forest stands depends on several factors. Mild winters, large-scale agriculture, which provides enough easily accessible attractive crops for the game, and improperly implemented hunting management—including sometimes inappropriate feeding and baiting—are the leading causes of the increase in game density. Subsequently, the over-density and the inappropriate age and sex structure of wild game populations can be considered the leading causes of forest damage. Also, increased game concentration due to predation pressure of large carnivores outside the vegetation period, excessive illegal disturbances by humans (e.g., collection of cast antlers), and local unavailability of suitable food in winter can be mentioned. Moreover, the fragmentation of habitats due to urbanization and the construction of highway and road infrastructure, without sufficient preference for preserving natural migration corridors or improper hunting management due to outdated legislation, is important [3–5].

Large herbivores living in the wild are declining in most countries, whereas the opposite trend prevails in central Europe [6]. The red deer population is growing significantly in some European countries [7], and in Slovakia, it has almost doubled over the past twenty years [8]. In the Central European area, the red deer is considered a representative of cloven-hoofed game and ruminants, which cause the most significant damage in forest ecosystems by stripping the bark and browsing the non-woody parts of plants, such as assimilation organs, buds, and twigs [7,9,10].

Bark stripping is characteristic of winter when natural food sources are limited. The cause of the damage in question is a lack of quality food and one-sided nutrition and often, also, a lack of minerals and trace elements [11–14]. By stripping the bark, red deer expose trees to fungal infections, which can lead to wood rot and stem deformation. Additionally, the weakened, rotting stem becomes more prone to breaking during storms [10]. The damage is subsequently reflected in the economic valuation. However, economic loss is just one of the multiple issues within these stands. A severe infection in multiple trees could lead to the collapse of the entire stand due to reduced resistance to both biotic and abiotic factors [15].

Damage caused by red deer, whose population density has a long-term upward trend, is an increasing problem, but ungulates also use migratory flexibility to optimize access to spatially and temporally variable resources across seasons [16,17]. One of the options for partially eliminating the mentioned negatives, and thus also the damage caused by wild animals to forest plants, is supplementing the nutrition of wild animals through suitable mineral licks, which compensate for the lack of necessary nutrients.

The results of using tannins in animal production published in the scientific literature point to their favorable properties and beneficial effects on farm animals. The effect of tannins on animals, their role in the food of farm animals, and their use in the food industry have been dealt with by several authors. These effects can be both positive and negative. They affect the digestibility of proteins and thereby the functioning of the digestive system [18,19].

According to hunting statistics, the number of red deer in Slovakia was estimated at 33 thousand individuals in 2000; in 2020, it increased to 75 thousand. The rising trend appeared despite the increased hunting limits for 1997–2022 [20]. These adverse developments ultimately mean that the pressure on forest crops increases yearly. The plan for deer hunting in 2022 was at the level of 64,756 deer, representing an annual increase of 11,089 individuals. The spring counts were reported as of 31 March 2022; 76,625 individuals exceeded the standard population size by up to 1.94 times [21].

A significant economic rise in animal damage is also recorded in forest stands managed by the Forests of the Slovak Republic, a state enterprise. For the evaluated period of 2017 to 2023, there was an increase from about EUR 0.5 to over EUR 2.5 million, of which almost EUR 0.62 million was bark stripping and EUR 1.86 million was browsing damage. In 2022, the total damage caused by wild animals in Slovakia was estimated at EUR 2.9 million, of which EUR 1.66 million accounts for damage to forestry, and EUR 1.24 million was the cost for damage to agriculture [21].

The main objective of this study was to experimentally validate the efficacy of Tani Liz, a natural preparation, as a potential biological shield for forest stands against animal damage. We aimed to test the hypothesis that a tannin-based mineral lick could decrease the intensity of deer-induced damage. Therefore, two similar forested areas in the Western Carpathians were selected. Both places had similar natural conditions, current states, species composition of the forest, and the extent of bark-stripping damage before the experiment started. Subsequently, we expected that the total area of the bark-stripping wounds in the location treated with tannin over the winter would not be significantly increased (or stay at a similar level) between sample measurements in the autumn of the previous year and consecutive sample measurements in the spring of the following year. In contrast, we hypothesized that the total area of bark stripping in the second nearby location left without treatment (as control) would be enlarged due to the cumulative nature and unchanged intensity of the damage.

2. Materials and Methods

2.1. Study Areas

Two sets of forest stands located in the territory of the University Forest Enterprise of the Technical University in Zvolen were selected as experimental areas to verify the effectiveness of tannin-based mineral licks (product from LOBENS.r.o. Dovalovo, SR) in reducing red deer damage. The forest administration chose two locations, named Budča and Makovisko. At Makovisko, only around 7 individuals are hunted annually, and in Budča, approximately 20 individuals per year. There are no mouflons or fallow deer. Wild boars have been affected by African swine fever. In Makovisko, there were approximately 90 deer over 2277 hectares, which means 4.0 deer per 100 hectares, in 2022/2023. In the Budča area, there were approximately 250 deer over 6056 hectares, which is 4.1 deer per 100 hectares, in 2022/2023. The selected localities/experimental areas are separated by the city of Zvolen and the highway, which prevented interactions between game populations. Both experimental areas are under the forest management of the same entity, which also holds the hunting rights. This unified management approach, encompassing silviculture, forest protection, and hunting management (including hunting methods, care, and feeding), ensured a controlled environment for the experiment.

When selecting the locations, the maximal possible match/similarity in basic environmental and vegetation parameters, such as altitude, tree composition, and age, was achieved (Table 1). In the selected localities, Budča and Makovisko, a circle of 1 km² was drawn. The stands within or intercepted by the circle were included in the evaluated territory (Figure 1). The intention was to capture the largest possible area potentially threatened by game and, at the same time, to eliminate the potential influence of the terrain on the movement of animals and the spatial distribution of damaged trees. Both areas were originally similarly exposed and randomly threatened by forest animals. This fact is supported by the damage size recorded by the initial sample measurement before the experimental application of tannin treatment (Table 2). In total, 17 stands with an area of 135.09 ha at Budča and 15 stands of 147.95 ha at Makovisko were included in the experimental area. This is less than the home range of a red deer, meaning the selected localities are places where deer migrate or reside. The tannin treatment was applied in the Budča location and, at this experimental area, we did not apply feeding with salt or any other prevention measure for the whole experimental winter season.

Table 1. Age and species composition of forest stands in experimental areas.

Budča				Makovisko			
Stand	Area [ha]	Age *	Tree Species	Stand Number	Area [ha]	Age *	Tree Species
407	7.1	5-15-35	<i>Fs, Aa, Pa, Cb, Aps</i>	153	14.9	10-20-40	<i>Fs, Pa, Fe, Aps, Ld</i>
408	9.08	5-25-35-170	<i>Fs, Pa, Qp, Aa, Ld, Aps</i>	154	12.41	20-35-160	<i>Fs, Fe, Pa, Aps, Ld</i>
409	17.38	5-10-15-25-40-170	<i>Fs, Aa, Tc, Cb, Aps, Apl, Pa, Ld</i>	155a	15.85	20-40-145	<i>Fs, Tc, Um, Apl, Pa, Ld</i>
410	13	5-15-40-60	<i>Fs, Aa, Tc, Cb, Apl, Fe</i>	155b	0.55	60	<i>Fs</i>
412a	8.13	5-20-35	<i>Fs, Aa, Ld, Qp, Pa, Cb</i>	156	17.67	15-25-35-150	<i>Fs, Pa, Aps, Fe, Ld</i>
413	7.88	5-30-125	<i>Fs, Cb, Aa, Qp, Aps, Pav, Fe</i>	157	14.62	5-15-20-30-40-150	<i>Fs, Pa, Ld, Fe</i>
518a	12.57	5-15-35-150	<i>Fs, Pa, Aa, Aps, Qp</i>	158	12.79	10-15-30-130	<i>Fs, Pa, Ld</i>
519	7.89	40-110	<i>Fs, Pa, Ld, Aps, Aa</i>	159	14.3	30-135	<i>Fs, Pa, Ld</i>
520	8.03	5-30	<i>Fs, Pa, Aa, Ld</i>	160a	11.94	10-30-40-155	<i>Fs, Pa, Aps, Ld, Bp</i>
522	11.42	10-35-160	<i>Fs, Pa, Aa, Aps,</i>	160b	1.69	60	<i>Fs</i>
523	7.49	30	<i>Fs, Pa, Ld, Aa</i>	161a	5.45	5-15-60-120	<i>Fs, Ld, Pa, Aps, Qp, Cb</i>
524a	5.22	35	<i>Fs, Pa, Aa, Ld</i>	161b	1.84	55	<i>Ag, Fs, Fe, Bp</i>
524b	2.97	15	<i>Fs, Pa, Aps, Aa</i>	164a	9.99	15	<i>Fs</i>
525	9.79	4-15-20-35-140	<i>Fs, Pa, Qp, Aps, Fe, Cb, Aa</i>	280a	4.72	50-190	<i>Qp, Fs, Cb</i>
527b	0.77	10	<i>Fs, Pa, Fe</i>	280c	9.23	40	<i>Qp, Cb, Fs, Aps, Tc, Ld</i>
528a	5.75	35-95	<i>Fs, Aa, Fe, Aps</i>	-	-	-	-
528b	0.62	10	<i>Fs, Fe, Aps</i>	-	-	-	-
Total	135.09	-	-	Total	147.95	-	-

Explanations: Aa—*Abies alba*, Ag—*Alnus glutinosa*, Apl—*Acer platanoides*, Aps—*Acer pseudoplatanus*, Bp—*Betula pendula*, Cb—*Carpinus betulus*, Fe—*Fraxinus excelsior*, Fs—*Fagus sylvatica*, Ld—*Larix decidua*, Pa—*Picea abies*, Pav—*Prunus avium*, Qp—*Quercus petaea*, Tc—*Tilia cordata*, Um—*Ulmus montana*. * The series of ages is provided for some stands if the stand is composed of tree cohorts/layers or spatial subparts of distinct ages. The labels after the numbers a, b, c are specific designations of parts of the forest. They are necessary in forestry maps in our region.

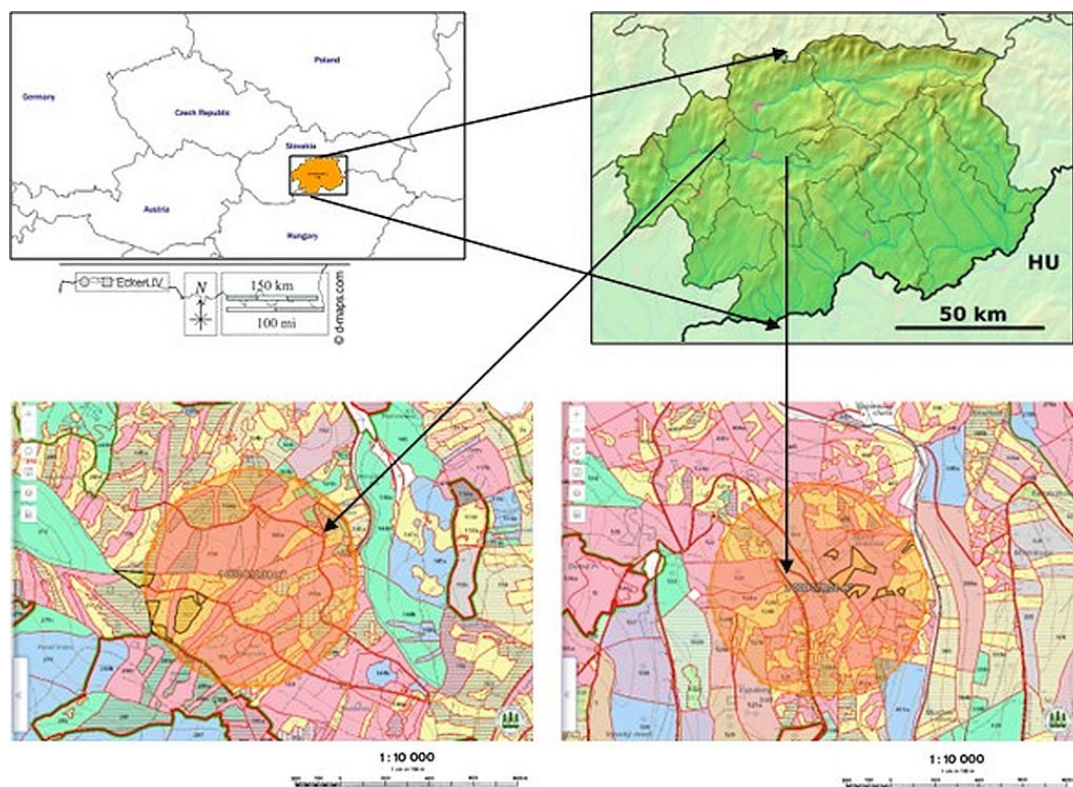
**Figure 1.** Location of experimental areas (on the left is the area of Budča in the Kremnica Mountains—the area with tannin licks; on the right is the control (untreated) area of Makovisko in the Javorie Mountains).

Table 2. Bark-stripping damage.

Locality	Budča		Makovisko	
Period	Autumn	Spring	Autumn	Spring
Number of plots (damaged/undamaged)	29/40	22/47	34/37	24/47
Mean damage per plot [m^2ha^{-1}]	41.66	27.26	42.11	84.19
The average size of the individual wound [m^2]	0.1204	0.0831	0.1236	0.2109
Total damage per experimental area [m^2]	5627	3682	6230	12,456

2.2. Sampling and Measurement

The representative plot samples were established in the forest cover. The plots were defined by a set of ten trees nearest to the sampling point. One sampling point and plot were established for every 2 ha of forest cover (systematic selection in every 200 m). A total of 140 plots were established (69 sampling plots were in Budča, and 71 were assessed at the Makovisko location). The total number of evaluated trees was 690 individuals at the Budča location and 710 at the Makovisko location, corresponding to 69 and 71 ten-tree sampling plots covering the experimental areas at a particular measurement time point.

Both areas were measured twice: in the autumn of 2022 and in the spring of 2023, before and after the experimental application of tannin licks at the selected location Budča. Thus, two independent representative plot samples were formed at two selected time points in both experimental areas (the research plots were not the same/identical during the measurements), and sample information about the bark-stripping extent was obtained before and after the winter season.

The tree species and the extent of damage by browsing or stripping were recorded for each sample tree. All the damage was included in the overall assessment, i.e., fresh, old, and repeated according to the methodology for detecting damage to forest stands by animals and damage assessment [22]. Only browsing and stripping damage caused by red deer (easily and unambiguously visually recognizable in the field) was recorded. The amount of browsing damage or stripping on individual trees was measured using a 20×20 cm plastic mesh with a mesh size of 5×5 mm. To facilitate the measurement, i.e., for accurate reading of the area of damage, 10×10 cm and 5×5 cm squares were highlighted on the measuring grid for more precise and faster reading of the size. The size of the wound was recorded to the nearest 1 cm^2 .

Also, the distance (spacing) of the third nearest neighbor from the central tree, a_3 , was measured at each plot (the central tree was the individual nearest to the sampling point). Based on this distance, the number of trees per 1 ha, $N.ha^{-1}$, was calculated according to the empirical formula at the sampling point/plot [23]:

$$N.h_a^{-1} = 11554.(\bar{a}_3)^{-1.99885} \quad (1)$$

Then, based on the number of trees per 1 ha and the size (area) of bark damage given in m^2 detected for 10 sample trees, the total bark damage per 1 ha in m^2 was easily calculated as plot damage times the number of trees per hectare divided by 10.

As already mentioned, two measurements were conducted in both considered experimental areas. The first evaluations of stripping damage by red deer were carried out simultaneously at the Budča and Makovisko locations in the second half of September 2022 (from 19 September to 29 September 2022). On 30 September 2022, after assessing the extent of damage to trees by stripping, mineral licks with tannin were distributed in the Budča area. At the control locality, licks with tannin were not placed. The second measurement of the stripping damage on a new independent sample of ten-tree plots was carried out in July 2023 (from 4 July 2023 to 14 July 2023).

It should be emphasized that the sampling plots in both monitoring periods were not identical, i.e., independent samples were formed. Placing the plots in the same spatial spots and remeasuring the identical tree sets was undesirable. The size of previous

damage almost surely affects the size of the following damage in the same set of trees. The probability of new damage occurrence on already damaged trees is surely lower than in undamaged tree sets, so providing reliable statistical proof of the positive effects of tannin licks in identical plots/tree sets would be far more difficult.

2.3. Applied Treatment

As mentioned, mineral licks of the trademark Tani Liz were distributed at the Budča location. Tani Liz is a pressed lick weighing 9 kg, intended for ruminants, produced by Product Feed Dunajska Streda. It is primarily designed to prevent metabolic disorders and improve the immunity of ruminants. The composition of the Tani Liz preparation is lignocellulose from *horse chestnut* (Farmatan Fiber preparation), sodium chloride, calcium carbonate, monocalcium phosphate, magnesium oxide, sodium sulfate, magnesium sulfate, water, and a premix of soil substances—the microelements calcium, sodium, sulfur, phosphorus, magnesium, selenium, zinc, cobalt, copper, manganese, and iodine. The preparation Farmatan Fiber is produced by the Slovenian company Tamin Sevnica.

The tannin is produced by hydrolysis of the bark and wood of sweet chestnut. The Tani Liz product comprises 10% of Farmatan Fiber tannin. Cloven-hoofed game consume licks with their tongues, where they perceive the astringent taste of tannin. Thus, its absorption is ensured through the salivary glands and digestion (according to the company's promotional leaflets). The active substances are castalagin and vescalagi, which have potent antimicrobial, antioxidant, antimutagenic, and antiparasitic effects in animals [24].

2.4. Statistical Analysis

A multifactorial analysis of variance (ANOVA) was used to demonstrate a statistically significant positive effect of tannin licks on the damage to the forest. In this analysis, the damaged area registered on individual sampling plots (only damaged non-zero plots were included) served as the dependent variable, and the experimental location (Makovisko/Budča) and the period of investigation (autumn 2022/spring 2023) and their interactions appeared as two independent factors. A non-parametric multiple rank comparison Tukey's HSD test was used to evaluate differences between pairs of levels for significant factors and their interactions in more detail.

The assumption of normal distribution of dependent variables was tested before the ANOVA application. The Shapiro–Wilks test result disagreed with this assumption (Figure A1). Therefore, a logarithmic transformation of the original values was performed (Figure A2), leading to a successful correction of non-normality (again approved by a Shapiro–Wilks test). Therefore, the multifactorial ANOVA design using the log-transformed values of the non-zero plot damaged areas was applied. In addition, descriptive information about the numbers of undamaged plots and the mean and total damage in distinct monitoring periods were provided to complete information about the effects of tannin treatments.

3. Results

The meaningfulness of using tannin-based mineral licks as a tool for forest protection was demonstrated (Table 2). After the second measurement, the total area damaged by bark stripping (calculated as location area in ha times the mean bark-stripping damage per plot in $\text{m}^2 \cdot \text{ha}^{-1}$) was $12,456 \text{ m}^2$ at the Makovisko location (without tannin), which is damage more extensive than in autumn (+100%). Although the number of plots with the occurrence of damaged trees decreased (from 34 to 24), the mean registered damage was doubled.

At the location of Budča, where licks with tannin were distributed, bark stripping of 3682 m^2 was recorded in spring, less than in autumn, whereas the number of damaged plots and mean damage per plot decreased simultaneously. Another interesting piece of evidence of increased/decreased intensity of damage provided information about the sizes of individual wounds; there were more and larger wounds in the Makovisko area.

Thus, the early indication was that the intensity of damage at Budča was not enlarged, at least. On the contrary, the situation at Makovisko suggests the continuation of very in-

tense damage to forest stands (despite a mild winter and more intensive shooting in comparison to Budča, as confirmed by personal communication with the hunting administration).

To confirm the early indications by statistical means and due to the sampling character of the survey, the results of the multifactorial ANOVA were provided in Tables 3 and 4.

Table 3. Analysis of variance for log-transformed values of bark-stripping damage (non-zero plots).

Parameter	SS	Degrees of Freedom	MS	F	p
Absolute value	1066.18	1	1066.18	760.123	0.00000
Location	0.592	1	0.592	0.4224	0.51717
Period	3.22	1	3.22	2.2954	0.13276
Location*Period	12.59	1	12.59	8.9758	0.00341
Error	147.277	105	1.403	-	-

Explanations: SS—sum of squares, PC—mean square, F—criterion, p—significance level; red probabilities are statistically significant at 0.05 level. * denotes a combination.

Table 4. Tukey’s honestly significant difference (HSD) test for unequal N (approximate probabilities for post hoc tests).

Location	Period	{1}—3.2679	{2}—2.9271	{3}—2.7280	{4}—3.7661
Budča	Autumn	-	0.77557	0.31062	0.46705
	Spring	0.77557	-	0.94438	0.09364
Makovisko	Autumn	0.310618	0.944375	-	0.01584
	Spring	0.467052	0.093637	0.01584	-

Note: the red probabilities are significant at a 5% significance level.

The ANOVA showed that individual factors do not significantly impact the mean plot damage area, but their interaction does. To further analyze this, multiple non-parametric comparisons of significant interactions were subsequently performed (Table 4). Here, the combination of “Makovisko” with “Spring” is significantly different from the combination of “Makovisko” with “Autumn”. Other interactions did not distinguish significantly and can be assessed as comparable.

The results are displayed in Figure 2, which contains the arithmetic means of log-transformed damage (mean of log-transformed bark-stripping areas registered on sampling plots recalculated to $\text{m}^2 \text{ha}^{-1}$) and 95% confidence limits for log-transformed means according to combinations/interactions of measurement periods and studied locations.

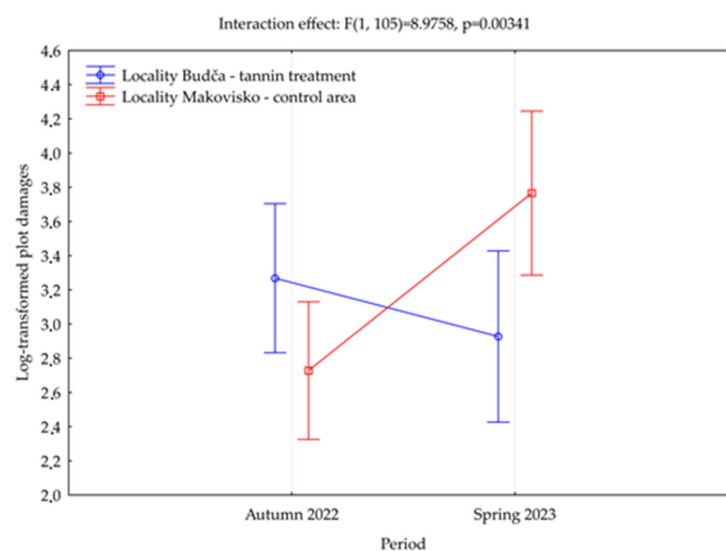


Figure 2. Interaction effects of studied locations and periods on log-transformed bark-stripping damage registered on damaged plots (95% confidence intervals for the mean of log-transformed values are displayed by whiskers).

In the Budča location, where the Tani Liz mineral licks were applied, the mean damage per sample plot did not significantly increase in winter, remaining at a similar level. This confirms an early indication that the new damage caused by bark stripping in the given location did not change significantly.

At the same time, we can see no significant differences among mean bark-stripping areas per plot between localities in autumn 2022 (start of the experiment), approving the expected homogeneity of natural conditions, state of the forest, game populations, and their hunting management at experimental areas at the start of the experiment.

Most importantly, the only significant difference in mean bark-stripping areas was recorded between measurement periods (autumn vs. spring) at the control location (Makovisko). The early indication provided in Table 3 was statistically confirmed also here—the intensity of the damage during the winter period was significantly enlarged at the control locality left without the treatment.

4. Discussion

The constantly increasing numbers of wild animals and the resulting damage to forest stands call for the need to search for new, effective, but at the same time ecological methods of forest protection. From an economic point of view, browsing and bark stripping/peeling by red deer is of the greatest importance in forest stands in Slovakia. In this context, the state of the deer population is currently quite alarming compared to the nutritional value of the hunting grounds. According to the principles of integrated forest protection, which is based on the concept of integrated pest management (IPM), the main emphasis should be placed on prevention. The implemented measures should not significantly affect the complex synecological relationships of the protected ecosystem [25,26]. The current development of products that eliminate game damage is aimed at replacing less effective, ecotoxicologically, and hygienically unsuitable chemical components with products based on natural substances [27]. The goal is to select plant extracts (terpenes, tannins, alkaloids, resins, etc.) with a strong effect on the taste and smell organs of animals, which will protect the trees against damage [28].

In this regard, we consider the application of tannin-based mineral licks among the new environmentally friendly methods with relatively high efficiency in eliminating bark-stripping damage, with a beneficial effect for the cloven-hoofed game itself. An essential component of these preparations is natural tannin, which supports digestion and immunity and helps defend against pathogens. The licks can act as antiparasitic, antibacterial, and antitumor, reducing pain. Animals often consume the bark and shoots of trees containing tannins due to dyspeptic problems. In the spring, young shoots contain excessive protein, which can cause stomach acidification. The rumen cannot process excessive protein evenly with other substances, whereas consumed tannins bind to proteins and slow digestion, relieving the animal [29].

Therefore, the replacement of natural sources of tannin with mineral licks has some potential to prevent browsing and bark-stripping damage. Another reason for the use of tannin-based licks is the fact that currently not a single preparation against browsing and bark peeling is registered in Slovakia [30]. The use of other mechanical methods of protection is limited by the size of the stems and is often time-, technically, and financially demanding [27].

The effects of tannins on the digestive tracts of livestock have recently been discussed by several authors [31–33], where their impact on health and vitality was evaluated predominantly positively. Addisu reported that none of the tannin sources tested negatively affected the animals' food consumption, growth, and behavior. Similarly, no undesirable effects were recorded during the additional observation of the red deer during and after our experiment. In general, the proper nutrition of animals and their supplementary feeding are frequent topics of scientific research (e.g., [34]). As already mentioned, the positive effects of tannin in herbivores have recently been addressed by several authors and works [18,32,35–44], but specifically utilizing tannin as a natural substance intended

for the prevention of bark-stripping damage caused by cloven-hooved animals has not yet been independently studied. The presented results can therefore be considered original in the field of integrated forest protection.

Based on the obtained results, we consider the use of tannin as a biological method to reduce bark-stripping/peeling damage to be a promising and efficient method, in which, compared to the standard methods used in Slovakia, such as reducing the numbers of wild animals by shooting, feeding, using fences and repellents, no significant negatives were recorded. For example, the installation of fences is generally one of the most effective methods of protection, but further large-scale fencing of new territories without the subsequent demanding removal of old equipment is unrealistic, uneconomical, and, in a forest environment, unaesthetic in the long term. Also, the feeding of wild game usually contributes to an increase in the concentration of animals in the vicinity of feedlots and feeding grounds, which triggers other interrelated factors that harm the surrounding components of the environment. Oversaturation of game with grain feed often causes digestive disorders [45].

Therefore, animals seek wood with tannin due to health problems (acidification of the stomach and antibacterial and antiparasitic effects) [38,46,47]. Sufficient relatively easily available tannin in licks can, according to our observations, contribute to a significant reduction in damage to forest stands. The effect of tannins on red deer seems to be relatively strong, so consuming them from mineral licks reduces the special nutritional needs of animals, which results in considerably eliminating the risk of consuming tree bark [48].

Although during the research, we did not notice any side effects of mineral tannins on animals, trees, and the surrounding environment, the professional literature also describes negative impacts depending on the type of tannin, the preparation, the composition, the amount of the dose, and the concentration of tannins [49]. Tannins bind to proteins in the rumen, thereby reducing their degradation, but at low protein and high fiber concentrations in the diet, they have a rather harmful effect. With their bacteriostatic and bactericidal effects, they can adversely affect the metabolism of animals at increased concentrations [50]. Also, frequent and long-term consumption of leaves with a high tannin content can cause toxicity in animals with a simple stomach [51].

On the contrary, if the concentration of raw proteins in the diet exceeds the needs of the animals, their performance and digestibility improve, and the breakdown of proteins itself is optimized [50]. Therefore, the elimination of the negative effects of tannins on animals consists mainly in the correct dosage, the selected type of tannin [44], and the appropriate time of their placement in the stands.

Moreover, some *in vivo* and *in vitro* studies on farm animals have shown that plants containing various secondary metabolites such as condensed tannins can be effective in integrated parasite control. For example, consumption of tannin-rich plants has been shown to reduce the abundance of third-stage nematode larvae and reduce worm fecundity in naturally infected goats. Also, lambs can perceive the antiparasitic effects of tannins and increase their preference for tannin-rich feed [52]. From this point of view, it would be appropriate to investigate the antiparasitic effects of tannins also on wild animals.

5. Conclusions

The growing damage caused by deer game is becoming one of the most significant problems in forestry in Slovakia, not only because of the current level of damage or destruction of forest stands (trees) but also because of the financial costs of classic preventive and suppressive measures. For this reason, the effective regulation of harmful agents and mitigation of damage by introducing innovative, environmentally friendly methods and procedures is highly demanded.

Our research suggests that tannin-based mineral licks could be a promising solution in preventing deer-induced bark-stripping/peeling damage of forest stands. The measurements and evaluations carried out statistically confirmed the hypothesis that a mineral lick based on tannins can preventively eliminate the extent of damage caused by deer.

At the experimental area with the distributed tannin licks, bark-stripping damage was significantly lowered during the winter than at the untreated control location.

Based on the obtained results, we consider the use of tannin as a promising biological method to reduce bark-stripping/peeling damage to be potentially an effective and efficient method with no significant negatives recorded. The elimination of the potential negative effects of tannins on animals indicated in the literature can be easily prevented by the correct dosage, the selected type of tannin, and the appropriate time of its placement in the stands. In this regard, future research is needed. Also, the repetition of field experiments in different regions, natural conditions, and hunting areas will be useful for increasing the degree of confidence about the positive relation between tannin protection and decreased bark-stripping damage in locations with high densities of red deer populations.

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Appendix A

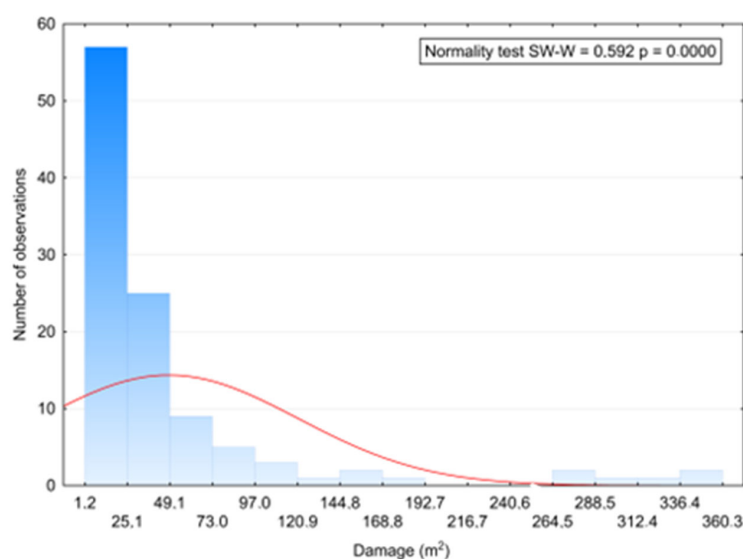


Figure A1. The distribution of the values of the dependent variable before the transformation showed that the normality requirement was not met (a Shapiro–Wilks test at 0.05 significance level was used).

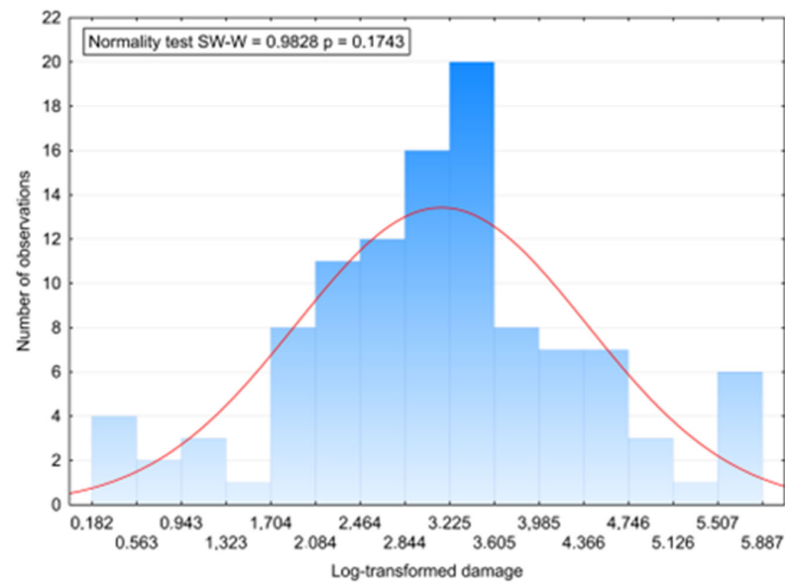


Figure A2. The distribution of the values of the dependent variable after the transformation showed that the normality requirement was met (a Shapiro–Wilks test at 0.05 significance level was used).



Figure A3. Damage to a tree caused by wild boars. The bark and wood are damaged by the rubbing of their bodies. Mud and boar hair remnants can be seen on the trunk. This kind of damage is present around the entire circumference of the trunk and up to a height of approximately 1 m.



Figure A4. Damage caused by deer antlers. This type of damage is characterized by remnants of bark that are either on the ground or hanging from the trunk. Often, there are also remnants of skin from the antlers.



Figure A5. Damage to trees caused by deer consuming the bark. This damage can reach up to a height of 2 m. Tooth marks are often visible (incisors). These tubes are wide and flat. Only the bark is damaged, not the wood.



Figure A6. Damage caused by a bear. Marks from sharp incisors and canines are visible. The bark is torn off and the wood is also damaged. The wood has been gnawed, and claw marks are visible. There may also be remnants of fur from the bear rubbing against the trunk. This damage is located high on the trunk.

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